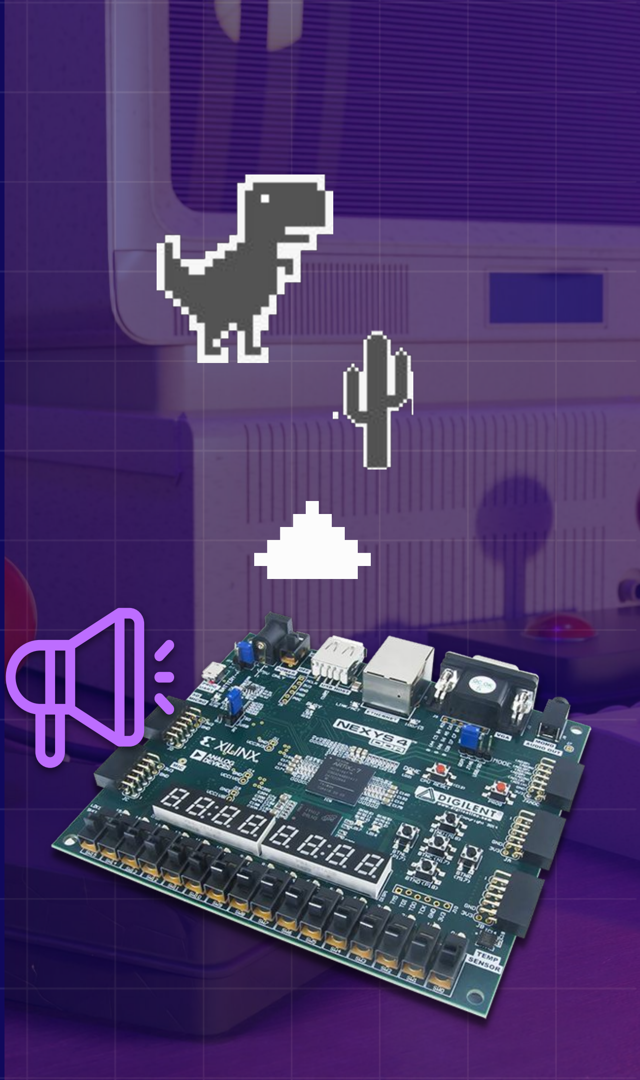
SHOUTMASTER 3000

An Arcade-Style Game of Decibel Control!

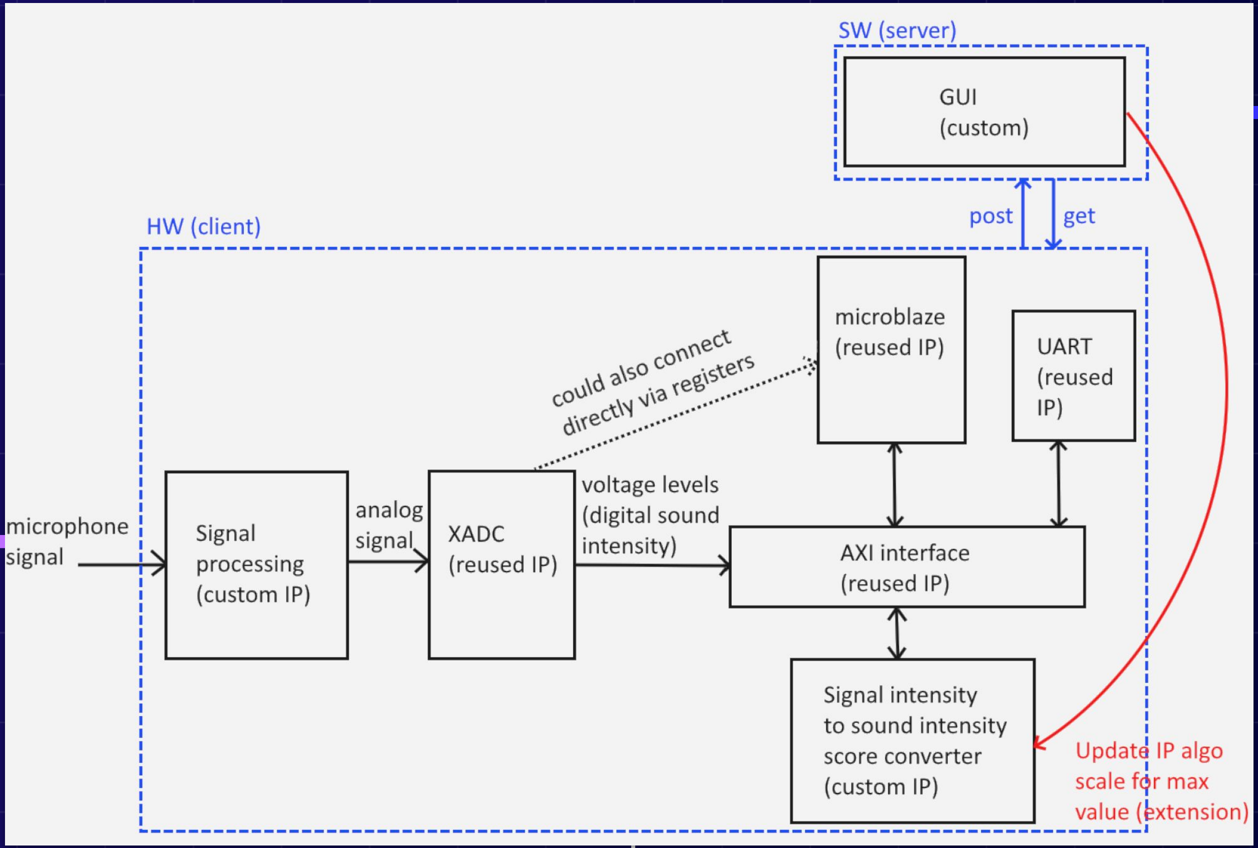
Group 23

High Level Description

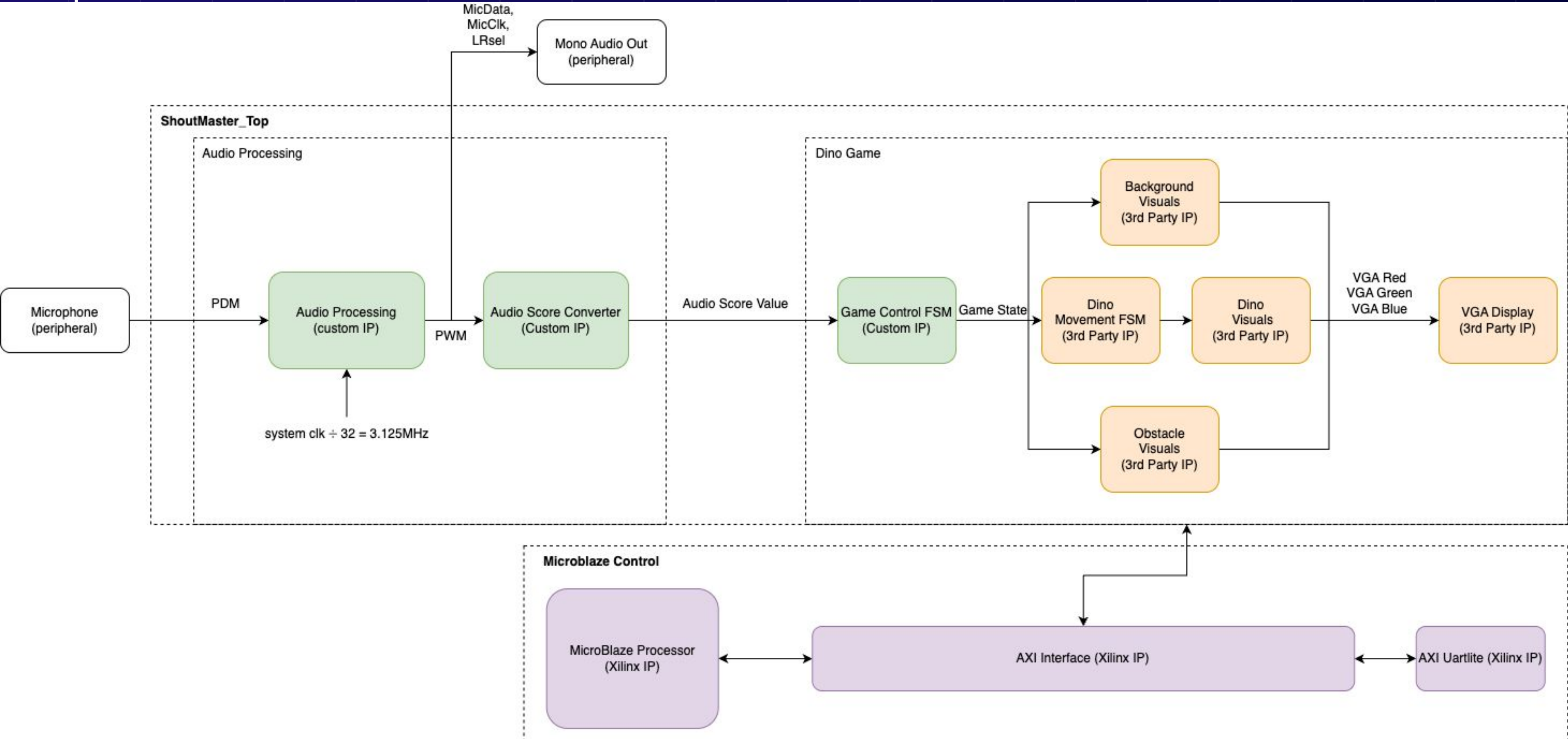
- Input: mic captures audio signal
- Outputs:
    - Processed sound to mono audio out
    - VGA displays dino game
- Dino game is audio-controlled
    - You shout, dino jumps
    - Game ends when dino hits obstacle



Initial Goals & Block Diagram



Final Implementation & Block Diagram



Major components #1:

DATA FLOW SCHEMA

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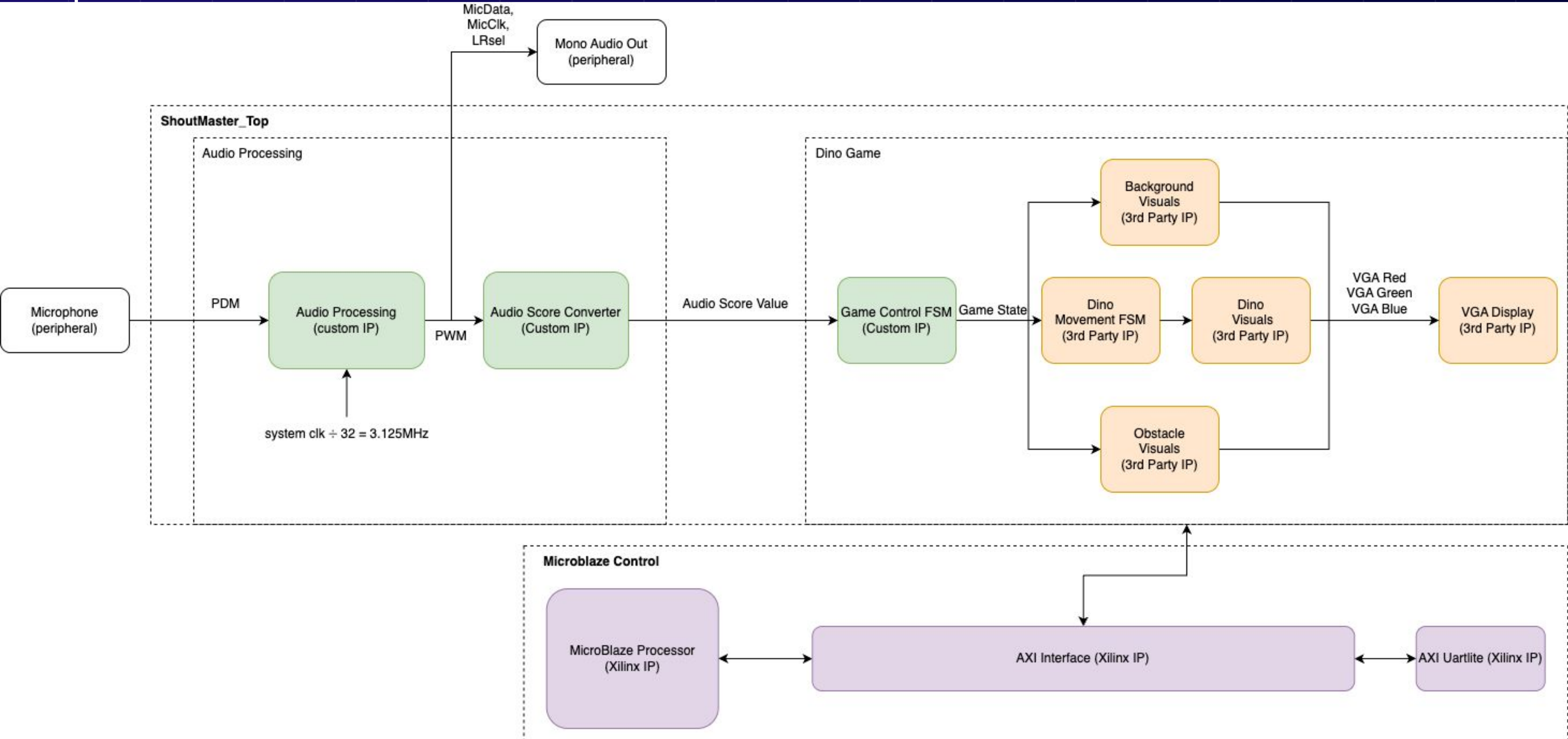


Major components #2:

CUSTOM IP

Audio Process IP

PDM mic in → PCM → PWM audio out

PDM mic in

Stream of 0/1

Density of 1s = amplitude

PCM

Sample PDM stream

Sampled value = amplitude

PWM audio out

Square wave

Duty cycle ∝ amplitude

Mic clk: sys_clk / 32 = 3.125MHz
Sample @ mic clk
Sample sets output hi/lo transition

Audio Score IP

Idea: volume-sensitive button

PWM accumulation

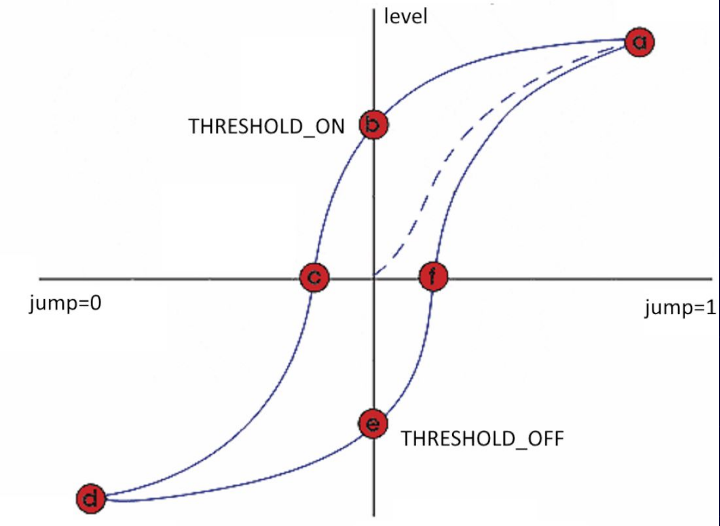
On every clock, if pwm is high, a fixed STEP is added to an internal sound counter

Decay

Each cycle, sound is also subtracted (baseline), simulating natural decay - prevent saturation

Hysteresis

Prevent flickering from noise



Dino IP

FSM:

Collision Detection

Time = 0: Detect overlaps between dino and obstacles

Time = 1: Hold to decrease buffer

Time = 2: Go to dead state

Motion (game state)

S1: Dino moves to right;

Background/Obstacles moves to left

S0/S2: Dino/Background/Obstacles stop moving

Major components #3:

PROCESSOR SYSTEM SOFTWARE

MicroBlaze Processor

- MicroBlaze functions as a score tracker
- Obstacle detection module counts jumps
- Count stored in AXI slave register
- MicroBlaze reads the value via AXI and sends it to SDK

PROJECT COMPLEXITY

On-board microphone	0.5 pts
On-board audio output port	0.5 pts
VGA Display without Microblaze	1 pt
GUI	0.75 pts
IP cores implemented in HW	0.5 pts
Total	3.25 pts

Improvements

Integrate Audio Response

- Add time-domain, frequency-domain data to game VGA display
- Visualize the user input

Increase Game Complexity

- Add more obstacles, i.e. birds, stones etc.
- Add fast, slow modes to the game

Demo

Thank you!